

Features to be Implemented in the Next Deliverable

Since this deliverable is the last phase of the Smart Course Companion project, all essential features specified in the project specifications have been effectively implemented. The system now has a completely integrated backend, user-specific data processing, permanent data storage, and full frontend and backend component interaction. As a result, there are no more essential features that need to be added within the parameters of this project.

However, if the system were to be expanded beyond the present deliverable, a number of new features and enhancements may be taken into consideration to improve the application's overall usefulness, usability, and scalability.

Adding a GPA estimate function to all courses is one possible improvement. A GPA calculator would offer a more thorough picture of a student's academic status, even though the existing system enables users to monitor individual course performance. This functionality might provide useful information for academic planning by dynamically calculating GPA based on course weights and grades kept in the system.

The option to export academic data in formats like PDF or CSV would be another beneficial feature. Users would be able to download and save their progress reports, grades, and course information using this feature. Additionally, it could be helpful for preserving backups outside of the system or exchanging academic data.

The system's utility would be greatly increased by incorporating automatic deadline reminders, especially via email alerts. The system would actively promote time management and lower the possibility of missing submissions by alerting users of impending tests or due dates.

More sophisticated data visualization techniques might be used for data representation. For instance, performance patterns over time, course comparisons, or assessment completion rates might all be shown using interactive charts or graphs. Users would be able to comprehend their academic progress more clearly and intuitively as a result.

The user interface and user experience (UI/UX) might be significantly enhanced from a design standpoint. Accessibility and user happiness would increase with features like support for both bright and dark modes, improved responsiveness across various screen sizes, and more sophisticated layouts.

Further development may also concentrate on improving role-based features for administrators or teachers. For example, more thorough statistics might be offered, such insights into student involvement or aggregated performance data. This would increase

the system's value for both instructional personnel and students.

Lastly, security and authentication system enhancements might be taken into account. Stronger password rules, session management, and data encryption are some further techniques that might improve the system's dependability and security even if the existing design guarantees that users can only view their own data.

In conclusion, even though the Smart Course Companion system currently satisfies all necessary project objectives, these potential features represent significant extensions that, if further development were to continue, could greatly improve its efficacy, scalability, and overall user experience.